

Task 1 - RAG Question Answering: One-Page Explanation

System: FastAPI + PyMuPDF + Gemini embeddings + ChromaDB + Gemini generation

1. Design parameter choice: chunk size and overlap

I treated chunking as a parameter to test instead of choosing it only by intuition. I compared 400/60, 800/120, and 1200/180 (chunk size/overlap in characters) on one controlled document with 10 manually labeled, paraphrased questions. The 400/60 configuration gave the strongest retrieval result in this diagnostic: 90% Hit@1 and 100% Hit@3. I therefore used a 400-character chunk with 60-character overlap in the final API, with top_k=3. This is a choice supported by this small evaluation, not a claim that the same values are universally optimal.

Chunk / overlap	Chunks	Hit@1	Hit@3
400 / 60	7	90%	100%
800 / 120	4	70%	90%
1200 / 180	3	80%	100%

2. Actually observed failure and its cause

With the 800/120 configuration, the question *"How frequently is dashboard information updated?"* missed the expected evidence even within the top three retrieved chunks, although the document explicitly says that dashboard data is refreshed every five minutes. The larger chunk grouped that fact with more surrounding content, making the specific evidence less focused in the chunk embedding and hurting its retrieval rank in this test. This failure was one reason I preferred the smaller 400/60 configuration.

3. Metric tracked and what it told me

I tracked retrieval Hit@1 and Hit@3. Hit@1 checks whether the expected evidence is in the first retrieved chunk; Hit@3 checks whether it appears anywhere in the three chunks that the application sends to the LLM. For 400/60, Hit@1 was 9/10 and Hit@3 was 10/10. This told me that the correct evidence was not always ranked first, but the top-three context contained it for every question in this diagnostic set. That supports using top_k=3 for the current implementation.

4. What is unfinished and what I would do next

The current evaluation is intentionally small: one controlled document and 10 questions. Scanned/image-only PDFs are also unsupported because OCR is not implemented, and chunking is character-based rather than document-structure-aware. Given more time, I would evaluate on a larger and more varied document set, compare sentence/section-aware chunking, measure answer faithfulness in addition to retrieval quality, add stronger document lifecycle management and integration tests, and add OCR for scanned PDFs.